

Hypothesis Test

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Motivation — A Guiding Example

Suppose we want to know if exposure to asbestos is associated with lung disease. We take rats and randomly divide them into two groups: one is exposed to asbestos, the other is not. We then compare the disease rate in the two groups.

- **Null hypothesis:** the disease rate is the **same** in the two groups.
- **Alternative hypothesis:** the disease rate is **not the same** in the two groups.

If the exposed group has a much higher rate of disease, we **reject** the null hypothesis and conclude that the evidence favors the alternative. This is an example of **hypothesis testing**.

The Testing Framework

Partition the parameter space Θ into two disjoint sets Θ_0 and Θ_1 , and test

$$H_0 : \theta \in \Theta_0 \quad \text{versus} \quad H_1 : \theta \in \Theta_1.$$

We call H_0 the **null hypothesis** and H_1 the **alternative hypothesis**.

Let X be a random variable with range $\mathcal{X}$. We test a hypothesis by finding a subset of outcomes $R \subset \mathcal{X}$ called the **rejection region**:

$$X \in R \implies \text{reject } H_0, \quad X \notin R \implies \text{retain (do not reject) } H_0.$$

Test Statistics and Critical Values

Usually the rejection region R has the form

$$R = \{ x : T(x) > c \},$$

where T is a **test statistic** and c is a **critical value**.

The problem of hypothesis testing is to find an appropriate test statistic T and an appropriate critical value c .

Warning

There is a tendency to use hypothesis testing even when it is not appropriate. Often, estimation and confidence intervals are better tools. Use hypothesis testing only when you want to test a well-defined hypothesis.

The Legal Trial Analogy — Two Types of Error

Hypothesis testing is like a legal trial: we assume someone is innocent unless the evidence strongly suggests guilt. Similarly, we retain H_0 unless there is strong evidence to reject it.

- **Type I error:** rejecting H_0 when H_0 is true.
- **Type II error:** retaining H_0 when H_1 is true.

	Retain H_0	Reject H_0
H_0 true	✓	type I error
H_1 true	type II error	✓

Summary of outcomes of hypothesis testing.

Definition — Power Function, Size, Level

The **power function** of a test with rejection region R is

$$\beta(\theta) = \mathbb{P}_\theta(X \in R).$$

The **size** of a test is

$$\alpha = \sup_{\theta \in \Theta_0} \beta(\theta).$$

A test has **level** α if its size is less than or equal to α .

The size is the largest probability of a type I error, taken over all $\theta \in \Theta_0$.

Simple vs. Composite; One- and Two-Sided Tests

- A hypothesis of the form $\theta = \theta_0$ is a **simple hypothesis**.
- A hypothesis of the form $\theta > \theta_0$ or $\theta < \theta_0$ is a **composite hypothesis**.

Two-sided test: $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$

One-sided test: $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$

$H_0 : \theta \geq \theta_0$ versus $H_1 : \theta < \theta_0$

The most common tests are two-sided.

Testing a Normal Mean (Part 1)

Let $X_1, \dots, X_n \sim N(\mu, \sigma)$ with σ known. We test

$$H_0 : \mu \leq 0 \quad \text{versus} \quad H_1 : \mu > 0,$$

so $\Theta_0 = (-\infty, 0]$ and $\Theta_1 = (0, \infty)$. Reject H_0 if $T = \bar{X} > c$.

Power function

$$\begin{aligned} \beta(\mu) &= \mathbb{P}_\mu(\bar{X} > c) = \mathbb{P}_\mu\left(\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} > \frac{\sqrt{n}(c - \mu)}{\sigma}\right) \\ &= \mathbb{P}\left(Z > \frac{\sqrt{n}(c - \mu)}{\sigma}\right) = 1 - \Phi\left(\frac{\sqrt{n}(c - \mu)}{\sigma}\right), \end{aligned}$$

where $Z \sim N(0, 1)$.

Testing a Normal Mean (Part 2)

$\beta(\mu)$ is increasing in μ , so

$$\text{size} = \sup_{\mu \leq 0} \beta(\mu) = \beta(0) = 1 - \Phi\left(\frac{\sqrt{n}c}{\sigma}\right).$$

For a size α test, set this equal to α and solve for c :

$$c = \frac{\sigma \Phi^{-1}(1 - \alpha)}{\sqrt{n}}.$$

Conclusion

We reject H_0 when $\bar{X} > \sigma \Phi^{-1}(1 - \alpha)/\sqrt{n}$; equivalently, when

$$\frac{\sqrt{n}(\bar{X} - 0)}{\sigma} > z_\alpha, \quad z_\alpha = \Phi^{-1}(1 - \alpha).$$

Most Powerful Tests

It would be desirable to find the test with the highest power under H_1 , among all size α tests. Such a test, if it exists, is called **most powerful**.

Finding most powerful tests is hard, and in many cases they do not even exist. Instead, we will study three widely used tests:

1. the **Wald test**,
2. the χ^2 **test** (Pearson's test for multinomial data),
3. the **permutation test**.

The Wald Test

Let θ be a scalar parameter, let $\hat{\theta}$ be an estimate of θ , and let $\widehat{\text{se}}$ be the estimated standard error of $\hat{\theta}$.

Definition — The Wald Test

Consider testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$. Assume $\hat{\theta}$ is asymptotically Normal:

$$\frac{\hat{\theta} - \theta_0}{\widehat{\text{se}}} \rightsquigarrow N(0, 1).$$

The size α **Wald test**: reject H_0 when $|W| > z_{\alpha/2}$, where

$$W = \frac{\hat{\theta} - \theta_0}{\widehat{\text{se}}}.$$

The Wald Test Has Asymptotic Size α

Statement

Asymptotically, the Wald test has size α ; that is,

$$\mathbb{P}_{\theta_0}(|W| > z_{\alpha/2}) \rightarrow \alpha \quad \text{as } n \rightarrow \infty.$$

Proof. Under $\theta = \theta_0$, we have $(\hat{\theta} - \theta_0)/\widehat{\text{se}} \rightsquigarrow N(0, 1)$. Hence the probability of rejecting when the null is true is

$$\mathbb{P}_{\theta_0}(|W| > z_{\alpha/2}) = \mathbb{P}_{\theta_0}\left(\frac{|\hat{\theta} - \theta_0|}{\widehat{\text{se}}} > z_{\alpha/2}\right) \rightarrow \mathbb{P}(|Z| > z_{\alpha/2}) = \alpha,$$

where $Z \sim N(0, 1)$. ■

Comparing Two Prediction Algorithms (Part 1)

Test algorithm 1 on a set of size m and algorithm 2 on an independent set of size n . Let X be the number of errors for algorithm 1 and Y the number for algorithm 2, with $X \sim \text{Binomial}(m, p_1)$ and $Y \sim \text{Binomial}(n, p_2)$.

To test $p_1 = p_2$, write $H_0 : \delta = 0$ versus $H_1 : \delta \neq 0$ where $\delta = p_1 - p_2$. The MLE is $\hat{\delta} = \hat{p}_1 - \hat{p}_2$, with estimated standard error

$$\widehat{\text{se}} = \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{m} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n}}.$$

The size α Wald test rejects H_0 when $|W| > z_{\alpha/2}$ where

$$W = \frac{\hat{\delta} - 0}{\widehat{\text{se}}} = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{m} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n}}}.$$

The power is largest when p_1 is far from p_2 and the sample sizes are large.

Comparing Two Prediction Algorithms — Paired (Part 2)

If we use the *same* test set for both algorithms, the two samples are no longer independent. Instead, let $X_i = 1$ if algorithm 1 is correct on case i , and $Y_i = 1$ if algorithm 2 is correct on case i . Define $D_i = X_i - Y_i$.

Case	X_i	Y_i	D_i
1	1	0	1
2	1	1	0
3	1	1	0
4	0	1	-1
5	0	0	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	0	1	-1

Let

$$\delta = \mathbb{E}(D_i) = \mathbb{P}(X_i = 1) - \mathbb{P}(Y_i = 1).$$

The plug-in estimate is $\hat{\delta} = \bar{D} = n^{-1} \sum_i D_i$ with $\widehat{\text{se}}(\hat{\delta}) = S/\sqrt{n}$, where $S^2 = n^{-1} \sum_i (D_i - \bar{D})^2$.

To test $H_0 : \delta = 0$ versus $H_1 : \delta \neq 0$, use $W = \hat{\delta}/\widehat{\text{se}}$ and reject if $|W| > z_{\alpha/2}$. This is a **paired comparison**.

Comparing Two Means

Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be two independent samples from populations with means μ_1 and μ_2 . Test the null hypothesis that $\mu_1 = \mu_2$, i.e. $H_0 : \delta = 0$ versus $H_1 : \delta \neq 0$ with $\delta = \mu_1 - \mu_2$.

The nonparametric plug-in estimate of δ is $\hat{\delta} = \bar{X} - \bar{Y}$ with estimated standard error

$$\widehat{\text{se}} = \sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}},$$

where s_1^2 and s_2^2 are the sample variances.

Size α Wald test

Reject H_0 when $|W| > z_{\alpha/2}$, where

$$W = \frac{\hat{\delta} - 0}{\widehat{\text{se}}} = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}}.$$

Statement

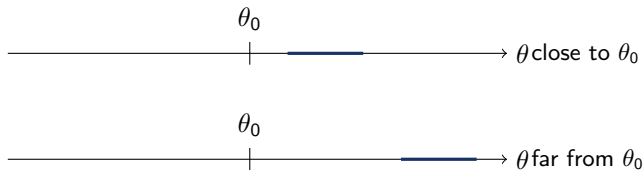
The size α Wald test rejects $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$ if and only if $\theta_0 \notin C$, where

$$C = (\hat{\theta} - \widehat{\text{se}} z_{\alpha/2}, \hat{\theta} + \widehat{\text{se}} z_{\alpha/2}).$$

Thus, testing the hypothesis is **equivalent** to checking whether the null value θ_0 lies inside the $1 - \alpha$ confidence interval.

Statistical vs. Scientific Significance

When we reject H_0 we say the result is **statistically significant**. But a result can be statistically significant while the size of the effect is small — not **scientifically** significant.



Both confidence intervals exclude θ_0 , so in both cases the test rejects H_0 . In the first case the estimate is close to θ_0 (little practical value); in the second it is far from θ_0 (scientifically meaningful). Confidence intervals are often more informative than tests.

Reporting “reject H_0 ” or “retain H_0 ” is not very informative. Instead, ask for *every* α whether the test rejects at that level. If the test rejects at level α , it also rejects at every $\alpha' > \alpha$; there is a smallest α at which the test rejects — the p -value.

Definition — p -value

Suppose that for every $\alpha \in (0, 1)$ we have a size α test with rejection region R_α . Then

$$p\text{-value} = \inf \{ \alpha : X^n \in R_\alpha \}.$$

That is, the p -value is the smallest level at which we can reject H_0 .

Interpreting p -values

p -value	evidence
$< .01$	very strong evidence against H_0
$.01 - .05$	strong evidence against H_0
$.05 - .10$	weak evidence against H_0
$> .1$	little or no evidence against H_0

Two warnings

- A **large** p -value is *not* strong evidence in favor of H_0 : it can occur because H_0 is true, or because H_0 is false but the test has low power.
- Do *not* confuse the p -value with $\mathbb{P}(H_0 \mid \text{Data})$. The p -value is **not** the probability that the null hypothesis is true.

Statement

Suppose the size α test rejects H_0 if and only if $T(X^n) \geq c_\alpha$. Then

$$p\text{-value} = \sup_{\theta \in \Theta_0} \mathbb{P}_\theta(T(X^n) \geq T(x^n)),$$

where x^n is the observed value of X^n . If $\Theta_0 = \{\theta_0\}$ then

$$p\text{-value} = \mathbb{P}_{\theta_0}(T(X^n) \geq T(x^n)).$$

In words: the p -value is the probability (under H_0) of observing a value of the test statistic the same as or more extreme than what was actually observed.

Computing the p -value — Intuition

Let $t_{\text{obs}} = T(x^n)$ be the value of the test statistic we actually observed.

1. The size α test rejects H_0 only when $t_{\text{obs}} \geq c_\alpha$.
2. A larger α means a bigger rejection region, so the cutoff c_α slides down toward t_{obs} . Turn α up from 0: the first time we can reject is when c_α just reaches t_{obs} .
3. That borderline α is the p -value — the smallest level at which the data reject H_0 .
4. At the borderline, $c_\alpha = t_{\text{obs}}$. Since α is always the size of the test,

$$p\text{-value} = \sup_{\theta \in \Theta_0} \mathbb{P}_\theta(T(X^n) \geq c_\alpha) = \sup_{\theta \in \Theta_0} \mathbb{P}_\theta(T(X^n) \geq T(x^n)).$$

In words

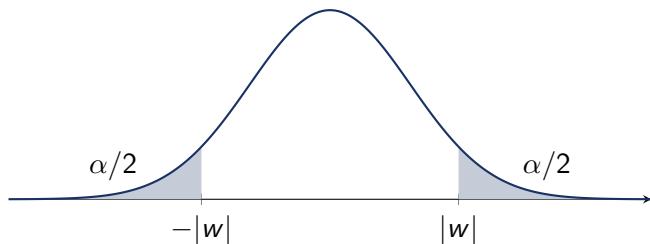
The p -value is the probability, under H_0 , of getting a test statistic at least as extreme as the one we observed.

The p -value for the Wald Test

Statement

Let $w = (\hat{\theta} - \theta_0)/\widehat{\text{se}}$ denote the observed value of the Wald statistic W . The p -value is

$$p\text{-value} = \mathbb{P}_{\theta_0}(|W| > |w|) \approx \mathbb{P}(|Z| > |w|) = 2\Phi(-|w|), \quad Z \sim N(0, 1).$$



The χ^2 Distribution

Let $Z_1, \dots, Z_k$ be independent standard Normals and let $V = \sum_{i=1}^k Z_i^2$. Then V has a χ^2 **distribution with k degrees of freedom**, written $V \sim \chi_k^2$, with density

$$f(v) = \frac{v^{(k/2)-1} e^{-v/2}}{2^{k/2} \Gamma(k/2)}, \quad v > 0.$$

It can be shown that

$$\mathbb{E}(V) = k, \quad \mathbb{V}(V) = 2k.$$

The **upper α quantile** is $\chi_{k,\alpha}^2 = F^{-1}(1 - \alpha)$, where F is the CDF; that is, $\mathbb{P}(\chi_k^2 > \chi_{k,\alpha}^2) = \alpha$.

Pearson's χ^2 Test for Multinomial Data

Pearson's χ^2 test is used for multinomial data. If $X = (X_1, \dots, X_k)$ has a Multinomial(n, p) distribution, then the MLE of p is

$$\hat{p} = (\hat{p}_1, \dots, \hat{p}_k) = (X_1/n, \dots, X_k/n).$$

Let $p_0 = (p_{01}, \dots, p_{0k})$ be a fixed vector. We want to test

$$H_0 : p = p_0 \quad \text{versus} \quad H_1 : p \neq p_0.$$

Definition

Pearson's χ^2 statistic is

$$T = \sum_{j=1}^k \frac{(X_j - np_{0j})^2}{np_{0j}} = \sum_{j=1}^k \frac{(X_j - E_j)^2}{E_j},$$

where $E_j = \mathbb{E}(X_j) = np_{0j}$ is the expected value of X_j under H_0 .

Each term compares an *observed* count X_j with its *expected* count E_j under the null hypothesis.

Statement

Under H_0 , $T \rightsquigarrow \chi_{k-1}^2$. Hence the test

$$\text{reject } H_0 \text{ if } T > \chi_{k-1, \alpha}^2$$

has asymptotic level α . The p -value is $\mathbb{P}(\chi_{k-1}^2 > t)$, where t is the observed value of the test statistic.

The p -value is the tail area of the χ_{k-1}^2 density to the right of the observed statistic t .

Mendel's Peas

Mendel's theory of inheritance predicts four types of progeny (round yellow, wrinkled yellow, round green, wrinkled green) with probabilities

$$p_0 = \left(\frac{9}{16}, \frac{3}{16}, \frac{3}{16}, \frac{1}{16} \right).$$

In $n = 556$ trials he observed $X = (315, 101, 108, 32)$, so $np_{01} = 312.75$, $np_{02} = np_{03} = 104.25$, $np_{04} = 34.75$.

$$T = \frac{(315 - 312.75)^2}{312.75} + \frac{(101 - 104.25)^2}{104.25} + \frac{(108 - 104.25)^2}{104.25} + \frac{(32 - 34.75)^2}{34.75} = 0.47.$$

The $\alpha = .05$ critical value for χ^2_3 is 7.815. Since $0.47 < 7.815$, we do **not** reject H_0 . The p -value is

$$p\text{-value} = \mathbb{P}(\chi^2_3 > .47) = .93.$$

The data do not contradict Mendel's theory.

The Permutation Test — Idea

The permutation test is a nonparametric method for testing whether two distributions are the same. It is **exact**: it is not based on large-sample approximations.

Suppose $X_1, \dots, X_m \sim F_X$ and $Y_1, \dots, Y_n \sim F_Y$ are two independent samples, and we test

$$H_0 : F_X = F_Y \quad \text{versus} \quad H_1 : F_X \neq F_Y.$$

Let $T(x_1, \dots, x_m, y_1, \dots, y_n)$ be a test statistic, for example

$$T(X_1, \dots, X_m, Y_1, \dots, Y_n) = |\bar{X}_m - \bar{Y}_n|.$$

The Permutation Distribution

Let $N = m + n$ and consider all $N!$ permutations of the data $X_1, \dots, X_m, Y_1, \dots, Y_n$. For each permutation, recompute the test statistic T ; denote these values $T_1, \dots, T_{N!}$.

Under H_0 , each of these values is equally likely. The distribution $\mathbb{P}_0$ that puts mass $1/N!$ on each T_j is the **permutation distribution** of T .

p -value

Let t_{obs} be the observed value of the test statistic. Assuming we reject when T is large,

$$p\text{-value} = \mathbb{P}_0(T > t_{\text{obs}}) = \frac{1}{N!} \sum_{j=1}^{N!} I(T_j > t_{\text{obs}}).$$

A Toy Permutation Test

Suppose the data are $(X_1, X_2, Y_1) = (1, 9, 3)$ and $T(X_1, X_2, Y_1) = |\bar{X} - \bar{Y}| = 2$.

permutation	value of T	probability
$(1, 9, 3)$	2	1/6
$(9, 1, 3)$	2	1/6
$(1, 3, 9)$	7	1/6
$(3, 1, 9)$	7	1/6
$(3, 9, 1)$	5	1/6
$(9, 3, 1)$	5	1/6

The p -value is $\mathbb{P}(T > 2) = 4/6$.

Algorithm for the Permutation Test

Algorithm

1. Compute the observed value of the test statistic $t_{\text{obs}} = T(X_1, \dots, X_m, Y_1, \dots, Y_n)$.
2. Randomly permute the data and compute the statistic again using the permuted data.
3. Repeat the previous step B times, giving $T_1, \dots, T_B$.
4. The approximate p -value is

$$\frac{1}{B} \sum_{j=1}^B I(T_j > t_{\text{obs}}).$$

Usually it is not practical to evaluate all $N!$ permutations, so we sample randomly. In large samples the permutation test gives results similar to large-sample tests, so it is most useful for **small** samples.

The Multiple Testing Problem

Sometimes we conduct many hypothesis tests. For example, a DNA microarray experiment with 2,638 genes would involve 2,638 separate tests.

Suppose each test is conducted at level α . For any *one* test, the chance of a false rejection is α . But the chance of *at least one* false rejection among many tests is much higher. This is the **multiple testing problem**.

We discuss two methods:

- the **Bonferroni method**,
- the **Benjamini–Hochberg (BH) method**.

The Bonferroni Method

Consider m hypothesis tests, H_{0i} versus H_{1i} for $i = 1, \dots, m$, and let $P_1, \dots, P_m$ be the corresponding p -values.

The Bonferroni Method

Given p -values $P_1, \dots, P_m$, reject null hypothesis H_{0i} if

$$P_i < \frac{\alpha}{m}.$$

Theorem

Using the Bonferroni method, the probability of falsely rejecting any null hypotheses is less than or equal to α .

The Bonferroni Method — Proof and Example

Proof. Let R be the event that at least one null hypothesis is falsely rejected, and let R_i be the event that the i -th null is falsely rejected. By the union bound,

$$\mathbb{P}(R) = \mathbb{P}\left(\bigcup_{i=1}^m R_i\right) \leq \sum_{i=1}^m \mathbb{P}(R_i) = \sum_{i=1}^m \frac{\alpha}{m} = \alpha. \blacksquare$$

Example

With $\alpha = .05$ and $m = 2,638$ genes, $.05/2638 = .00001895375$. Hence, for any gene with p -value less than $.00001895375$, we declare a significant difference. The Bonferroni method is very **conservative**.

The False Discovery Rate

The Bonferroni method tries to make it unlikely that you make *even one* false rejection. A less conservative idea is to control the **false discovery rate (FDR)**.

	H_0 not rej.	H_0 rej.	Total
H_0 true	U	V	m_0
H_0 false	T	S	m_1
Total	$m - R$	R	m

Outcomes in multiple testing.

The **False Discovery Proportion** is

$$\text{FDP} = \begin{cases} V/R & \text{if } R > 0 \\ 0 & \text{if } R = 0, \end{cases}$$

the proportion of rejections that are incorrect. Then

$$\text{FDR} = \mathbb{E}(\text{FDP}).$$

The Benjamini–Hochberg (BH) Method

The BH Method

1. Let $P_{(1)} < \dots < P_{(m)}$ denote the ordered p -values.
2. Define

$$\ell_i = \frac{i \alpha}{C_m m}, \quad R = \max \{ i : P_{(i)} < \ell_i \},$$

where $C_m = 1$ if the p -values are independent and $C_m = \sum_{i=1}^m (1/i)$ otherwise.

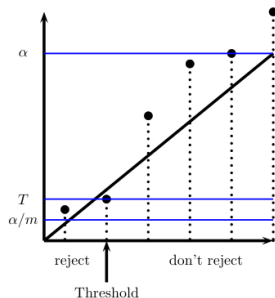
3. Let $T = P_{(R)}$, the **BH rejection threshold**.
4. Reject all null hypotheses H_{0i} for which $P_i \leq T$.

Theorem (Benjamini and Hochberg)

If this procedure is applied, then regardless of how many nulls are true,
 $\text{FDR} = \mathbb{E}(\text{FDP}) \leq (m_0/m) \alpha \leq \alpha$.

Comparing the Methods

Six ordered p -values $P_{(1)} < \dots < P_{(6)}$ plotted against their ranks; the upward line has slope α/m ($\ell_i = i\alpha/m$).



- **Uncorrected** ($P_i < \alpha$): the four smallest p -values.
- **Bonferroni** ($P_i < \alpha/m$): no rejections.
- **BH**: threshold $T = P_{(R)}$ is the last p -value under the sloping line ($R = 2$) — two hypotheses rejected.

A Worked Comparison

Suppose 10 independent hypothesis tests give the ordered p -values

0.00017, 0.00448, 0.00671, 0.00907, 0.01220,
0.33626, 0.39341, 0.53882, 0.58125, 0.98617.

With $\alpha = 0.05$ and $m = 10$:

- **Bonferroni:** reject any hypothesis with p -value less than $\alpha/10 = 0.005$. Thus only the **first two** hypotheses are rejected.
- **BH:** find the largest i with $P_{(i)} < i\alpha/m$. Here

$$i = 5 : \quad P_{(5)} = 0.01220 < \frac{5(0.05)}{10} = 0.025,$$

and $P_{(6)} = 0.33626 \not< 0.030$. So $R = 5$ and we reject the **first five** hypotheses.